Rafael A. Rodriguez-Sanchez

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About

I'm a PhD researcher interested in how agents can build useful, abstract representations of the world to support planning and hierarchical RL. My work focuses on hierarchical reinforcement learning, representation learning, and world-modeling, where I develop energy-based, contrastive, and action-driven representation learning algorithms. I enjoy working across the stack — from theory and algorithm design to training systems using PyTorch, JAX, and distributed multi-GPU pipelines. I also have hands-on experience with LLM finetuning (SFT and RL methods).

Technical Skills

Languages: Python, MATLAB — **Frameworks:** PyTorch, JAX, purejaxrl — **Topics:** Reinforcement Learning, Representation Learning, Contrastive Learning, Hierarchical RL, Energy-based Models, World Models, Diffusion Models, Latent Variable Models, Variational Inference — **Systems & Training:** Distributed Training, Multi-GPU Training, SLURM, Experiment Tracking, Scalable ML Systems — **Math:** Optimization, Linear Algebra, Probability Theory, Variational Principles

Professional Experience

Amazon — Applied Scientist Intern, Dialog Research Cambridge, MA
Jun 2021 – Sep 2021

- Research finetuning strategies (SFT and RL using PyTorch) of large language models (LLMs) to map natural language instructions to Alexa API calls improving performance over legacy rule-based systems.

Politecnico di Milano, AIRLab — Research Fellow Milan, Italy
Oct 2018 – Jul 2019

- Designed and evaluated computer vision tracking algorithms for intelligent defense simulations, improving flare detection robustness using OpenCV and MATLAB.

STMicroelectronics — Research Development Intern Agrate-Brianza, Italy
Feb 2015 – Jul 2015

- Optimized embedded C algorithms for real-time image acquisition in wearable devices, increasing throughput and reducing latency.

Education

Brown University — Intelligent Robot Lab Providence, RI
Ph.D. in Computer Science, Expected Dec 2025
Advisor: George Konidaris
Dissertation: *Action-driven Algorithms for Representation Learning in Reinforcement Learning*

Politecnico di Milano — AI Robotics Lab Milan, Italy
M.S. in Computer Science and Engineering, Cum Laude (Oct 2018)
MAECI Scholarship
Thesis: *A Variational Approach to Transfer Value Functions in Reinforcement Learning*

Selected Publications

- **From Pixels to Factors: Learning Independently Controllable State Variables for RL**. R. Rodriguez-Sanchez, C. Allen, G. Konidaris. Under review, 2025.

- **Intrinsically Motivated Discovery of Temporally Abstract Graph-based Models of the World**. A. Bagaria, A. De Mello Koch, R. Rodriguez-Sanchez, S. Lobel, G. Konidaris. RLC 2025.
- **Learning Abstract World Models for Value-preserving Planning with Options**. R. Rodriguez-Sanchez, G. Konidaris. RLC 2024.
- **RLang: A Declarative Language for Describing Partial World Knowledge to RL Agents**. R. Rodriguez-Sanchez*, B. Spiegel*, J. Wang, R. Patel, S. Tellex, G. Konidaris. ICML 2023.
- **Transfer of Value Functions via Variational Methods**. R. Rodriguez-Sanchez*, A. Tirinzoni*, M. Restelli. NeurIPS 2018.

Workshops

- Disentangling Independently Controllable Factors in RL. R. Rodriguez-Sanchez, C. Allen, G. Konidaris. NYRL Workshop 2025.
- Learning Abstract World Models for Value-preserving Planning with Options. R. Rodriguez-Sanchez, G. Konidaris. GenPlan @ NeurIPS 2023 (Contributed Talk).
- On the Relationship Between Structure in Natural Language and Models of Sequential Decision Processes. R. Rodriguez-Sanchez, R. Patel, G. Konidaris. LaRel @ ICML 2020.